

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 4 -- Answer Key

Q1 Solution (30 points)

a) E[X] and E[R]:

$$\begin{aligned}E[X] &= E[U(0)] + E[U(R)] = 30 + 25 = 55 \text{ minutes per cycle} \\E[R] &= E[50 * U(0)] = 50 * E[U(0)] = 50 * 30 = 1500 \text{ dollars per cycle}\end{aligned}$$

b) Long-run revenue rate:

$$\begin{aligned}\text{Long-run rate} &= E[R] / E[X] = 1500 / 55 = 300/11 \sim 27.27 \text{ dollars/minute} \\ \text{Per hour: } &27.27 * 60 \sim 1636.4 \text{ dollars/hour}\end{aligned}$$

c) E[X^2] and E[B]:

Cycle $X = U(O) + U(R)$, with $U(O)$ and $U(R)$ independent:

$$\begin{aligned}\text{Var}[X] &= \text{Var}[U(0)] + \text{Var}[U(R)] \\ \text{Var}[U(0)] &= (40-20)^2 / 12 = 400/12 = 100/3 \sim 33.33 \\ \text{Var}[U(R)] &= (1/\mu)^2 = 25^2 = 625 \quad [\text{Exp}(1/25) \text{ has Var} = \text{mean}^2] \\ \text{Var}[X] &= 100/3 + 625 = 100/3 + 1875/3 = 1975/3 \sim 658.33 \\ E[X^2] &= \text{Var}[X] + (E[X])^2 = 1975/3 + 55^2 = 1975/3 + 3025 \\ &= 1975/3 + 9075/3 = 11050/3 \sim 3683.3 \\ E[B] &= E[X^2] / (2 * E[X]) = (11050/3) / (2 * 55) = 11050 / 330 \sim 33.5 \text{ min}\end{aligned}$$

A random passenger arriving at the bus stop expects to wait about 33.5 minutes for the next departure -- more than $E[X]/2 = 27.5$ minutes due to the Inspection Paradox.

Q2 Solution (35 points)

a) M/M/inf stationary distribution:

$$\begin{aligned}\rho &= \lambda/\mu = 6 / (1/0.5) = 6 / 2 = 3 \\ \pi(n) &= e^{(-\rho)} * \rho^n / n! = e^{(-3)} * 3^n / n! \\ &[\text{Poisson}(3)] \\ \pi(0) &= e^{(-3)} \sim 0.0498 \\ \pi(3) &= e^{(-3)} * 27 / 6 = e^{(-3)} * 4.5 \sim 0.2240\end{aligned}$$

b) E[N] and E[T]:

$$E[N] = \rho = 3 \text{ patients} \quad (\text{mean of Poisson}(3))$$

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$$E[T] = E[N] / \lambda = 3 / 6 = 0.5 \text{ hours} = 30 \text{ minutes} = 1/\mu$$

$E[T] = 1/\mu$ exactly: patients spend ZERO time waiting (infinite servers $\Rightarrow$ immediate service), so the entire stay is treatment time.

c) Comparison with M/M/1 ($\mu' = 7$):

$$\rho' = \lambda/\mu' = 6/7 \sim 0.857$$

$$E[N'] = \rho'/(1-\rho') = (6/7)/(1/7) = 6 \text{ patients}$$

$$E[T'] = E[N']/\lambda = 6/6 = 1 \text{ hour} = 60 \text{ minutes}$$

M/M/inf ($\mu=2$): $E[T] = 30$ minutes

M/M/1 ($\mu'=7$): $E[T'] = 60$ minutes

M/M/inf gives half the waiting time despite having a slower server, because infinite servers eliminate queuing entirely.

Q3 Solution (35 points)

a) De Moivre's martingale:

$$r = q/p = (1/3)/(2/3) = 1/2$$

$$M(n) = r^{S(n)} = (1/2)^{S(n)}$$

$$\begin{aligned} E[r^{X(n+1)}] &= p \cdot r^{+1} + q \cdot r^{-1} = (2/3) \cdot (1/2) + (1/3) \cdot (2) \\ &= 1/3 + 2/3 = 1 \quad (\text{OK}) \Rightarrow M(n) \text{ is a martingale} \end{aligned}$$

b) Hitting probability h via OST:

$$r^4 = E[M(0)] = E[M(T)] = r^{10} \cdot h + r^0 \cdot (1-h)$$

$$(1/2)^4 = (1/2)^{10} \cdot h + 1 \cdot (1-h)$$

$$1/16 = h/1024 + 1 - h$$

$$h - h/1024 = 1 - 1/16$$

$$h \cdot (1023/1024) = 15/16$$

$$h = (15/16) \cdot (1024/1023) = 15360/16368 = 960/1023 \sim 0.9384$$

The upward-biased gambler ($p=2/3$) has about a 93.8% chance of reaching $N=10$ before ruin, starting at $k=4$.

c) $E[T]$ via Wald's equality:

$$E[X] = p - q = 2/3 - 1/3 = 1/3 \quad (\text{mean step size, upward drift})$$

$$E[S(T)] = 10 \cdot h + 0 \cdot (1-h) = 10h = 10 \cdot (960/1023) = 9600/1023 \sim$$

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9.384

Wald's equality: $E[S(T)] = E[X] * E[T]$

$$E[T] = E[S(T)] / E[X] = (9600/1023) / (1/3) = 28800/1023 \sim 28.15 \text{ steps}$$